

ChinookAnalytics: Analysis Objectives

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Abstract

This document defines the analytical objectives for ChinookAnalytics, derived from the Data Discovery findings of the curated **core** schema and aligned with typical business decision needs for a digital music retail operation.

Objectives are expressed as stakeholder questions and translated into measurable metrics, clear output structures, and reusable analytical views (marts) that enable reporting and insights.

1 Context from Data Discovery

The dataset represents a simplified digital music store where customers purchase tracks through invoices. The discovered structure supports robust attribution across time, geography, customers, and catalog dimensions.

Key structural notes from profiling:

- Revenue is represented by `invoice.total`; line-level economics are represented by `invoiceline.unit_price` and `quantity`.
- Referential integrity is complete along the backbone: `customer` → `invoice` → `invoiceline` → `track`, and attribution chain: `track` → `album` → `artist`, plus `track` → `genre` and `track` → `mediatype`.
- Known limitations:
 - churn/retention is weak (all customers have repeated purchases);
 - promotion impact is not measurable (no discount variables);
 - basket scaling via quantity is not present (quantity constant = 1).

2 Analytical Themes (Stakeholder View)

The objectives below answer six stakeholder decision themes:

- **Revenue Health:** Is the business growing, stable, or declining?
- **Market Strategy:** Which countries matter most, and how diversified are markets?
- **Customer Economics:** Who drives revenue, and how concentrated is dependency?

- **Catalog Strategy:** Which artists and genres generate most value?
- **Invoice Structure:** What explains high-value invoices and revenue composition?
- **Dependency Risk:** How fragile is revenue if top contributors change?

3 Metric Conventions

3.1 Time Bucketing

All time reporting uses:

`month_date = date_trunc('month', invoice_date)::date` and `year_date = date_trunc('year', invoice_date)::date`

3.2 Revenue and Units

```
invoice_revenue = core.invoice.total
line_revenue = core.invoiceline.unit_price × core.invoiceline.quantity
units =  $\sum$  core.invoiceline.quantity
```

3.3 Shares and Top-N Concentration

For an entity group g :

$$\text{revenue_share}(g) = \frac{\text{revenue}(g)}{\text{revenue}(\text{ALL})}$$

Top-N share:

$$\text{topN_share} = \frac{\sum_{g \in \text{TopN}} \text{revenue}(g)}{\sum_g \text{revenue}(g)}$$

3.4 Growth Rates

Month-over-month:

$$\text{mom_growth} = \frac{rev_t - rev_{t-1}}{rev_{t-1}}$$

Year-over-year:

$$\text{yoy_growth} = \frac{rev_t - rev_{t-12}}{rev_{t-12}}$$

If denominator is NULL/zero, return NULL.

3.5 Percentiles

Percentiles use:

$$P_q = \text{percentile_cont}(q) \text{ within group (order by } x)$$

Required percentiles: $q \in \{0.25, 0.50, 0.75, 0.90, 0.95\}$.

4 Objective 1: Revenue Health Over Time

4.1 Stakeholder Questions

- How does revenue evolve monthly and yearly?
- Are there consistent seasonal patterns?
- Is invoice value increasing or decreasing over time?
- Are revenue changes smooth or volatile?

4.2 Core Outputs (Model of Answer)

The analysis should produce:

- a monthly time series of revenue and invoice volume;
- a yearly summary with growth indicators;
- a volatility view (e.g., month-to-month revenue variation);
- average invoice value trend.

4.3 Analytical View

`marts.mart_revenue_time` (grain: month_date)

4.4 Required Metrics

- `invoices = COUNT(*)`
- `revenue = SUM(total)`
- `avg_invoice_value = AVG(total)`
- `mom_revenue_growth` using `LAG(revenue)`
- `yoy_revenue_growth` using `LAG(revenue, 12)`

5 Objective 2: Market Performance and Geographic Strategy

5.1 Stakeholder Questions

- Which billing countries generate most revenue and invoices?
- Which markets have the highest value per customer?
- Is revenue geographically diversified or dependent on few markets?

5.2 Core Outputs (Model of Answer)

The analysis should produce:

- ranking of countries by revenue and invoices;
- revenue share by country (concentration view);
- revenue per customer by country to compare market value.

5.3 Analytical View

`marts.mart_market_performance` (grain: `billing_country`)

5.4 Required Metrics

- `invoices = COUNT(*)`
- `revenue = SUM(total)`
- `avg_invoice_value = AVG(total)`
- `customers = COUNT(DISTINCT customer_id)`
- `revenue_per_customer = revenue / customers`
- `revenue_share = revenue / SUM(revenue)`

6 Objective 3: Customer Economics and Revenue Concentration

6.1 Stakeholder Questions

- Who are the highest-value customers (LTV)?
- How concentrated is revenue among customers?
- What does the customer value distribution look like?

6.2 Core Outputs (Model of Answer)

The analysis should produce:

- customer-level LTV table with invoice counts and dates;
- distribution summary (percentiles of LTV);
- Top-N customer revenue shares (Top 10/20/30).

6.3 Analytical View

`marts.mart_customer_economics` (grain: `customer_id`)

6.4 Required Metrics

- `invoices = COUNT(*)`
- `ltv = SUM(total)`
- `avg_invoice_value = AVG(total)`
- `first_purchase_date = MIN(invoice_date)`
- `last_purchase_date = MAX(invoice_date)`

6.5 Required Summary Queries

- LTV percentiles: $P_{25}, P_{50}, P_{75}, P_{90}, P_{95}$
- Top customer shares: Top-10, Top-20, Top-30 by ltv

7 Objective 4: Catalog Strategy (Artists and Genres)

7.1 Stakeholder Questions

- Which artists generate the most revenue and units?
- Which genres generate the most revenue and units?
- Is catalog revenue diversified or concentrated?

7.2 Core Outputs (Model of Answer)

The analysis should produce:

- artist ranking by revenue and units with revenue share;
- genre ranking by revenue and units with revenue share;
- concentration summary (Top 1/5/10 shares) for artists and genres.

7.3 Analytical Views

`marts.mart_portfolio_artist` (grain: `artist_id`)

`marts.mart_portfolio_genre` (grain: `genre_id`)

7.4 Required Metrics (Both Views)

- `units =  $\sum$  quantity`
- `revenue =  $\sum$ (unit_price  $\times$  quantity)`
- `revenue_share = revenue / total_revenue`

8 Objective 5: Invoice Structure and Revenue Composition

8.1 Stakeholder Questions

- What characterizes high-value invoices?
- Are high totals explained by more lines or more diversity (artists/genres)?
- How does invoice composition vary across countries or customers?

8.2 Core Outputs (Model of Answer)

The analysis should produce:

- invoice-level table with composition variables (lines and diversity);
- invoice total percentiles and a definition of “high-value” invoices;
- Top-10 highest invoices with decomposition fields.

8.3 Analytical View

`marts.mart_revenue_composition` (grain: `invoice_id`)

8.4 Required Metrics

- `invoice_total = core.invoice.total`
- `lines = COUNT(*)` invoice lines
- `units =  $\sum$  quantity`
- `distinct_tracks = COUNT(DISTINCT track_id)`
- `distinct_artists = COUNT(DISTINCT artist_id)`
- `distinct_genres = COUNT(DISTINCT genre_id)`
- `distinct_media_types = COUNT(DISTINCT media_type_id)`

8.5 Required Summary Queries

- Invoice total percentiles: $P_{25}, P_{50}, P_{75}, P_{90}, P_{95}$
- Top-10 invoices by `invoice_total` with: `lines, distinct_artists, distinct_genres`

9 Objective 6: Business Dependency and Concentration Snapshot

9.1 Stakeholder Questions

- How dependent is revenue on top artists?
- How dependent is revenue on top countries?
- How dependent is revenue on top customers?

9.2 Core Outputs (Model of Answer)

The analysis should produce a single executive summary row that reports:

- Top 1/5/10 artist revenue shares
- Top 1/5/10 country revenue shares
- Top 10/20 customer revenue shares

9.3 Analytical View

`marts.mart_dependency_summary` (grain: `snapshot_date`)

9.4 Required Metrics

- `total_revenue`
- `top1_artist_share`, `top5_artist_share`, `top10_artist_share`
- `top1_country_share`, `top5_country_share`, `top10_country_share`
- `top10_customer_share`, `top20_customer_share`

10 Deliverables Summary

Objective	Primary Analytical View(s)
Revenue over time	<code>mart_revenue_time</code>
Market performance	<code>mart_market_performance</code>
Customer economics	<code>mart_customer_economics</code>
Artist performance	<code>mart_portfolio_artist</code>
Genre performance	<code>mart_portfolio_genre</code>
Invoice composition	<code>mart_revenue_composition</code>
Dependency snapshot	<code>mart_dependency_summary</code>